

WASP-52b

R-0495

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230917 · created 2026-07-20T12:47:16+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460204.8705 +/- 0.0025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.014
Transit depth (Rp/Rs)^2	2.76 +/- 0.48 [%]
Orbital Inclination (inc)	89.73 +/- 1.27
Airmass coefficient 1 (a1)	0.991 +/- 0.018
Airmass coefficient 2 (a2)	0.006 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.81 %
Best Comparison Star	#2 - [403, 167]
Optimal Aperture	4.03
Optimal Annulus	11.44
Transit Duration (day)	0.0875 +/- 0.0018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.166 $\pm$ 0.014 vs 0.1646 (deviation 0.1 σ)
Duration fit vs published	0.0875 d vs 0.0758 d (deviation 6.48 σ)
Mid-time O-C	-13.4 min
Depth SNR	11.9
Residual scatter	1.81 %

Light curve

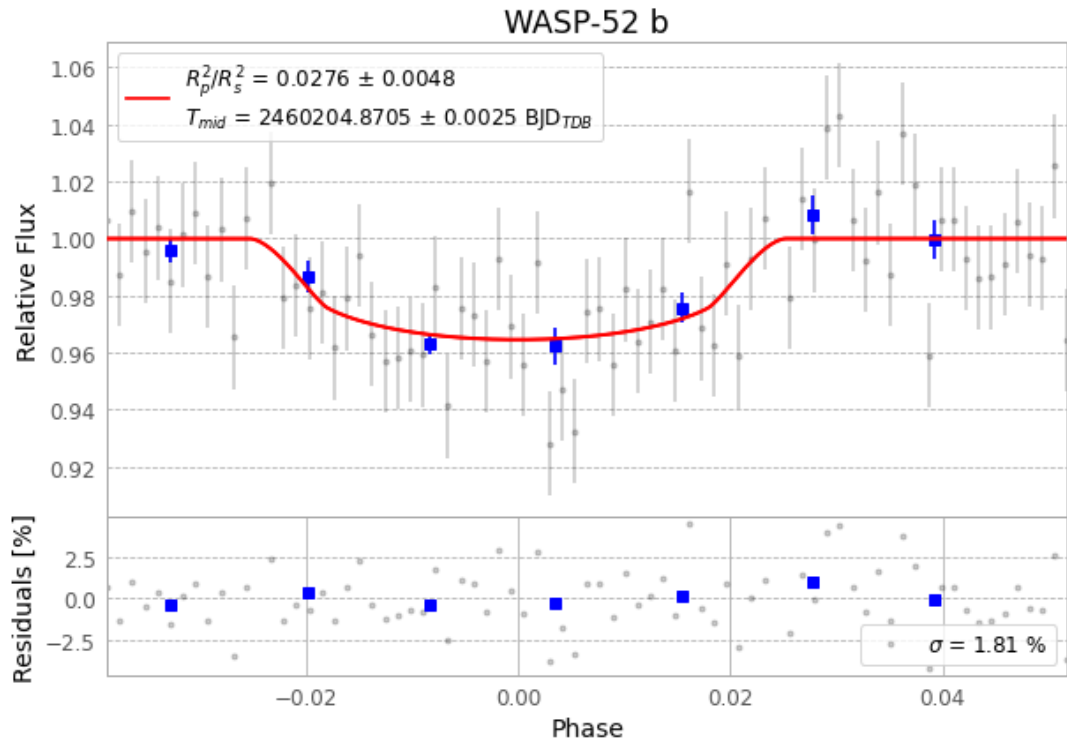


Figure 1 — FinalLightCurve_WASP-52 b_2023-09-17.png (SHA-256 f62a26cbce45f6b3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [293, 189], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0f6b33177b833f25...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

77 FITS frames (51 MB), WASP-52230917070616.FITS → WASP-52230917105415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230917.log (7d76c357dd97...), FinalLightCurve_WASP-52 b_2023-09-17.png (f62a26cbce45...), FinalLightCurve_WASP-52 b_2023-09-17.csv (cef47a7983dc...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 12:47Z.